

Limits Digest

Definition

$\lim_{x \rightarrow a} f(x) = L \Leftrightarrow$ for every $\varepsilon > 0$, δ exists where
if $0 < |x - a| < \delta$ then $|f(x) - L| < \varepsilon$.

Properties

$$\lim_{x \rightarrow a} x = a$$

$$\lim_{x \rightarrow a} C = C$$

$$\lim_{x \rightarrow a} C f(x) = C \cdot \lim_{x \rightarrow a} f(x)$$

$$\lim_{x \rightarrow a} (f(x) + g(x)) = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x)$$

$$\lim_{x \rightarrow a} (f(x) \times g(x)) = \lim_{x \rightarrow a} f(x) \times \lim_{x \rightarrow a} g(x)$$

$$\lim_{x \rightarrow a} \frac{1}{f(x)} = \frac{1}{\lim_{x \rightarrow a} f(x)}$$

$$\begin{cases} \lim_{x \rightarrow a} (f(g(x))) = f\left(\lim_{x \rightarrow a} g(x)\right) \\ \text{if } f(x) \text{ is continuous there} \end{cases}$$

$$\lim_{x \rightarrow a} (f(x))^n = \left(\lim_{x \rightarrow a} f(x)\right)^n$$

$$\lim_{x \rightarrow a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \rightarrow a} f(x)}$$

where $n \in \mathbb{N}_{>0}$

and if n is even, positive root is assumed

Inequalities

$$f(x) \lesseqgtr g(x) \text{ near } x = a$$

$$\Rightarrow \lim_{x \rightarrow a} f(x) \lesseqgtr \lim_{x \rightarrow a} g(x)$$

The Squeeze Theorem

If $f(x) \leq g(x) \leq h(x)$ near $x = a$,

and $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x)$,

then $\lim_{x \rightarrow a} g(x) = \lim_{x \rightarrow a} f(x)$.

Infinity Terminology

“Infinite Limit” $\Leftrightarrow$ vertical asymptote.

$$\lim_{x \rightarrow a} f(x) = \pm\infty$$

“Limit at Infinity” $\Leftrightarrow$ horizontal asymptote.

$$\lim_{x \rightarrow \pm\infty} f(x) = a$$

“Infinite Limit at Infinity” $\Leftrightarrow$ no asymptote.

$$\lim_{x \rightarrow \pm\infty} f(x) = \pm\infty$$

TODO

- indeterminate forms
- towards 0/inf patterns
- one-directional limits
- higher-d limits